Integer multiplication beyond the old frontier

Current CrocSwap proposals, evidence boundaries, and the next experiments

Source snapshot: 9 October 2026, 01:01 UTC / 03:01 CEST (Poland).
Selected immutable source versions were inspected to complete this review.

Executive decision

The public proposal frontier has crossed $\kappa = 10^{-4}$. The final refresh found PR #120 at $\kappa = 1.0911630664 \times 10^{-4}$, above #118 at $1.08542805968 \times 10^{-4}$. Both are drafts. PR #117 reports 1.044939×10^{-4} and a completed full repository rerun. The reviewed main branch still selects $5.1016920170078 \times 10^{-5}$. A draft claim, an author-reproduced result and a maintainer-reviewed checkpoint are different evidence categories. [S1–S3, S27]

> Final-refresh correction: #120 already combines #117's small complex DAG with #114's nondegenerate-subspace placement. Do not propose that combination as a new result. Use it as a baseline; investigate different complements, global target-chain placement and complete-moment objectives beyond its current greedy policy. [S5, S12, S13, S27]

The decisive change is no longer a better coordinate permutation in the old h23/h25 compiler. PR #104 changes residual compilation and its paid stopping interface; #97/#110 supply deferred schedules; #117 makes the complex producer substantially smaller. The current bottleneck is again the complex side, now near 1.09140×10^{-4}, with a stronger binary supplier already available. [S3, S6–S10, S27]

What this report establishes

The review compares the most recent 47 PR descriptions (#74–#120), the reviewed main checkpoint, selected earlier verification work, and pinned source code for the leading constructions. It independently recomputes the moment inequalities of the supplied #118 and #120 complex histograms with exact rational enclosures. It does not regenerate their graphs, certify the complete physical words, or prove the all-size multiplication theorem.

Recommended portfolio

GPU host: one controlled optimization lane beyond #120's already combined construction, with exact CPU-side verification and GPU discovery only where a measured batch implementation helps. **CPU breakthrough campaign:** retain broad independent research, but treat 10^{-4} as a public comparison milestone already claimed; use 2×10^{-4} as a structural planning target, not a promised outcome. Do not interrupt either campaign merely to reproduce every new headline.

This document proposes experiments only. No repository branches, workflows, running processes, or agent instructions were modified. The report and its small independent arithmetic check were generated separately.

1. Current leaders: score and evidence

All values below are conditional exponent savings in $T(n) = O(n(\log n)^{(1-\kappa)})$, not measured speedups. The numerical ordering uses the inspected versions, including later updates to older PRs. “Full” means author-reported repository reproduction; it is not an independent full rerun performed for this report. [S1–S7, S13–S17, S27]

PR / role	κ / 10^{-5}	Mechanism and present evidence
#120 — highest claim	10.911630664	Smaller #117 DAG + #114 saturated deferral. Draft; independent reflected-word checks reported; full rerun pending.
#118	10.85428059680	Smaller #117 complex DAG + deferred readouts. Draft; full rerun and independent reflected-word audit pending.
#117 — full-rerun claim	10.44939	28,705-role complex producer + stopped round-seven bit word. Non-draft; full Linux rerun reported.
#116	9.8699350338	Cube recurrence + balanced pairs + deferral. Draft; finite signed/reflected checks reported; full rerun pending.
#114 — updated head	9.8019072765	Core-aware cyclic producer + radical-complement deferral. Draft; newer-head CI pending in the description.
#113	9.39074	Cyclic-strip complex graph + deferral. Draft; expanded scalar charge corrected; full coverage pending.
#115	9.26336	Cube producer, backward frame enlargement and partial source gauges. Focused finite verification; unchanged broad suites not rerun.
#110	8.58478	Deferred readouts on both stopped networks. Original complex reflected schedule remains a written dependency.
#111	8.578151200855	Cyclic interval strips on both axes. Focused checks; full verification pending in the inspected body.
#112	8.0325171	Signed dependence reclamation. Complex primitive stronger than final κ because the retained bit side binds.
#109	7.9961717358	Weighted matching with unchanged role count. Full repository rerun and 42 checks reported.
#104 — architecture	7.79476	Stopped product-ring interchange + rational centers. Foundational interface change; not the numerical leader.
Reviewed main	5.1016920170078	#62 + #57 construction and #61 parameters. Maintainer-reviewed checkpoint; still conditional.

Do not compare against stale titles or summaries. For example, #116’s “4.17409% above #114” uses #114’s older $9.4744627892 \times 10^{-5}$ value. Against the inspected updated #114, the gain is approximately 0.69403%. PR #74 has also changed research families since its earlier aligned-frame result. Pin commits, not PR numbers alone. #119 ($9.42802139 \times 10^{-5}$) is closed, unmerged, and excluded from the open ranking; its access-order audit remains useful evidence. [S4, S5, S18, S28]

2. Why the frontier moved

From local profile tuning to a different compiler

The older aligned-frame track reduced roles and improved ordered rank profiles by changing graph partitions, coordinate flags, live controls and paid reclamation. Representative later values remain around 5.28×10^{-5} (#91–#106). Those mechanisms are useful, but their original objectives are not automatically relevant after a residual is charged as one child of its whole rank. [S19, S20]

PR #97 integrated Swapnil Jain's deferred round-seven bit word and a signed two-stage complex schedule with the community's physical and analytic interfaces. It explicitly accounts for nonzero auxiliary gauges, reflected continuity, commuting fan ordering, and a signed complex endpoint correction. The characteristic-two bit argument cannot simply replace this complex signed calculation. [S7]

The structural role of #104

PR #104 changes the recursive primitive. A common rational basis and opposite bank orders turn each proper projector residual of rank r into one reversed rank-r child. The proof must normalize the full involution, including complementary diagonal blocks, not merely factor its off-diagonal corner. Adapters execute on fixed-width atoms and are paid; an ordinary wrapper surrounds the entire stopped recursion. [S6, S8, S9]

```
a_bit = (1 − θ) a_coarse + θ a_old,     θ = 1/1000
```

The adapter term imposes θ > a_bit in that construction. The wrapper's constant number of calls is outside the recursion and must not be multiplied into every recursive child count. On the complex side, all 24 disjoint retained centers use exact division by 21 and a shared odd-denominator grid. These are new proof obligations compared with simply reusing the old fixed complex certificate. [S9, S10]

How #117, #118 and #120 exploit it

With the stopped round-seven bit supplier, the ordinary binary saving is about $1.240189553 \times 10^{-4}$. Improving it alone does not increase the current final κ because the complex stage binds. #117 instead reduces the complex producer from 44,918 roles in #104 to 28,705. #118 retains that graph and adds 3,020 deferred readouts. #120 uses saturated placement to retain 4,560. All three retain 28,705 complex roles; the deferred versions change the complete paid width profile, not that role count. [S2, S3, S11, S27, S29]

Dependency chain	Contribution to the current candidate
#36 → #104	Ordinary leaf; stopped atom interface; rational centers and one-child accounting.
Swapnil round 7 → #97 → #110 / #117	Deferred binary word; signed/reflected interface integration; stronger stopped bit supplier.
#117 → #118	Small complex DAG, then deferred readouts on that DAG.
#117 + latest #114 → #120	Saturated subspace placement is now already integrated; further experiments must go beyond this baseline.

3. Independent arithmetic and remaining headroom

For this review, a separate Python implementation evaluated the supplied #118 and #120 complex child histograms. It does not import their moment code. Inputs were transcribed from pinned JSON and checked for positive widths, strict contraction and exact rank mass. This checks supplied arithmetic, not the derivation of the physical network. [S11, S29]

$$H(s) = \Sigma_t\, n_t \cdot t/(mW) \cdot \exp(s \log(m/t))$$

Quantity	Recomputed value
Complex dimension m / width W	576 / 124,390,992
Rank mass $\Sigma_t\, t\, n_t$	71,647,349,312
Deficit $mW - \Sigma_t\, t\, n_t$	1,862,080 = N − L
Largest child	574 < 576
#118: b = 108566486 / 10^{12}	$1 - H(b) > 1.2888 \times 10^{-13}$
#118: b + 10^{-12}	$H(b + 10^{-12}) - 1 > 1.1055 \times 10^{-13}$
#120: b = 109140237 / 10^{12}	$1 - H(b) > 2.3781 \times 10^{-13}$
#120: b + 10^{-12}	$H(b + 10^{-12}) - 1 > 3.6107 \times 10^{-16}$

The logarithm enclosure uses power-of-two range reduction and 40 terms of the positive atanh series with an explicit geometric tail. The exponential enclosure uses a degree-nine Taylor polynomial with an explicit tail bound. All operations use rational numbers and outward rounding to a 2^{-180} grid. Thus the accepted and rejected grid points are separated rigorously. The complete source, transcription and rational endpoints accompany this report.

The available gain is mostly structural

For #120, b = $1.09140237 \times 10^{-4}$ and κ = $1.0911630664 \times 10^{-4}$. Their relative difference is only about 0.02193% and is not all freely recoverable: it includes strict assembly margins. Sharper certificate arithmetic cannot explain a major next jump. [S27, S29]

$$a_bit = 1240189553 / 10^{13} = 1.240189553 \times 10^{-4}$$

$$\kappa < a_bit / (1 + a_bit) \approx 1.24003576506 \times 10^{-4}$$

The second expression is an optimistic ceiling with this bit supplier and the retained balanced relation—not a universal complexity lower bound. It leaves roughly 13.64% above #120 if the complex side and all other requirements can be improved sufficiently. A target around 1.2×10^{-4} fits inside that numerical headroom. A target of 1.25×10^{-4} already needs a better bit supplier or a changed transfer.

For $\kappa = 2 \times 10^{-4}$ in the same scheme, a must exceed $1/4999 \approx 2.00040008 \times 10^{-4}$. At $\beta = 10^{-6}$, b must exceed approximately $2.00040208 \times 10^{-4}$, plus every other margin. Those necessary component gains are about 61.3% on the current bit saving and 83.3% on the current complex saving. This is a structural target, not a parameter sweep.

4. Priority E1: go beyond saturated placement

Important: the simple port is no longer new. The initial code comparison showed that #118 rejected degenerate target intersections while the latest #114 could retain a nondegenerate part. During final review, #120 appeared with exactly that integration. It reports a 0.52836% improvement over #118. Treat the port as baseline reproduction, not a proposed novel result. [S5, S12, S13, S27]

What has already been integrated

The #114 routine extracts diagonal-one pivots or hyperbolic pairs from the candidate intersection, discards its radical, and re-establishes containment after every extraction. It chooses candidates using dimension squared divided by the number of reached targets. #120 applies saturated placement to #117's fixed 28,705-role DAG and retains 4,560 deferred roles. [S13, S27]

The new package also reports an independent exported-word audit: 277,799 events per invocation per stage, complemented time reversal, bank exchange, zero violations and five rejected mutations. This closes a reported finite reflection gap still pending for #118. Full repository verification nevertheless remains pending in #120. We have independently checked its supplied moment, not rerun that word audit. [S2, S27, S29]

Four useful next variants

Variant	Change from pinned #120
A — exact control	Reproduce the published saturated placement and full paid profile.
B — alternate complements	Vary the exact nondegenerate complement, not just its dimension; recheck every target chain.
C — downstream cost	Replace dimension/target-count priority with the actual cost of current and future target-chain transitions.
D — local joint selection	Optimize a bounded conflict component of candidate frames together, then replay the complete word.

Freeze the scalar DAG, binary supplier and stopping parameters initially. Nondegenerate complements need not be unique: two choices of equal dimension may interact differently with future intersections and readouts. Record the retained Gram rank, targets reached, newly introduced transitions and the complete H(s). More deferred roles or dimensions alone do not establish an improvement.

Acceptance and continuation

Require independent exact subspace checks, all target containments and reflected incidences, correct signs, endpoint copies and expanded scalar readout charges. Never count radical dimensions as a valid nondegenerate frame. Compare against #120, not #118; reproducing the latter's already surpassed value is not a new result.

Suggested first budget: one control and three isolated four-worker jobs, within memory limits. Prototype complement choices on small exact instances before full DAG runs. Continue when a changed complete profile improves the moment or a scoped obstruction indicates a useful alternative. These are proposed experimental budgets, not measured runtimes.

5. E2–E3: optimize the complete physical program

E2. Global, moment-aware deferral selection

Beyond #120's saturated greedy policy, the next lever is joint placement rather than another arbitrary ordering. A role's useful deferral can be blocked by overlaps with many target chains. Build the role–target conflict structure, optimize small connected components exactly, and compare several deterministic global heuristics. Begin with bounded branch-and-bound, integer programming or explicit local exchanges; no claim of global optimality is needed.

Use the actual incremental cost at a fixed trial saving, not dimension or cardinality alone. If an entrance of rank r loses d dimensions and the actual exterior child grows from e to e+d, the two affected terms suggest the local change below. Use the actual e (often m−h in this family), not an assumed complement m−r. This is only a screen: target chains, copies, scalar work and other endpoints remain paid. [S2, S12, S28]

```
Δ = (r−d)^p + (e+d)^p − r^p − e^p,
p = 1−s, with normalization and multiplicity restored.
```

For $0 < p < 1$, concentrating a fixed rank mass into a larger child can reduce this concave cost; that explains the attraction of deferral. It does not establish that every legal deferral improves the whole word. Audit the largest child and row reserves whenever it approaches its parent dimension.

Test and stop rule: keep the bit supplier fixed; compare complete complex profiles around $b = 1.09140237 \times 10^{-4}$ and a research target near 1.2×10^{-4}. Expand only if the profile gain survives actual reflection and scalar charging. Do not spend a long search optimizing a surrogate after exact scoring repeatedly reverses its ranking.

E3. Signed reclamation and producer synthesis

PR #112 is useful even though its final κ is below the leaders: its complex supplier uses signed duplicate/sum/difference relations and 7,569 paid reclamations, while its weaker retained bit supplier limits the final result. Test whether those signed linear-signal techniques can be applied to #117's or #120's actual complex program. This is a new integration question, not a direct import of #57's binary XOR rules. [S16]

A second branch may vary the complex DAG itself: pair-star trees, cube constructions, balanced coarse sums and core-aware assembly can create different legal lifetimes. #116 is a caution: the original #115 graph with the deferral compiler lost, while a particular combined recurrence improved the full moment. Lower role count is neither a sufficient comparison nor a guarantee that two gains compose. [S4, S14]

Minimum evidence: exact signed scalar identities, independent live-signal and retirement tracking, all frame raises, the literal inverse/sign-negate/bank-swap program, and complete paid residuals. Track W, H(s), expanded scalar charge and actual runtime separately. Preserve #117's saved DAG as an immutable replayable input; do not claim to regenerate its original search merely by loading that DAG.

6. Structural CPU research: beyond the new cap

E4. Connect the new compiler to genuinely different geometry

The current RaD breakthrough branch already contains more than a frozen-profile obstruction. At checkpoint 8135f5e4 it records odd-weight kernels, exact dyadic center bases, cancellation-allowing scalar circuits, local noncommuting Clifford components and paid transport boundaries. It still makes no accepted $\kappa \geq 10^{-4}$ claim. Treat these as reusable components and scoped negatives—not as blank research directions that have never been tried. [S21]

In particular, the five-subset circuit has exact scalar improvements, but the retained monotone physical realization has zero rank deficit. A scalar cost of about 10.81 SSA roles per source at h20 is not a native-network certificate. The earlier 103.33-role tree and several stationary-frame shortcuts are already poor starting points. The next question is whether a different shared-release topology preserves a real benefit after all external frame transitions are paid. [S21]

A promising cross-check is to ask precisely which new residuals satisfy #104's one-child hypotheses. A proper projector residual in a valid common basis may admit that compiler; a noncommuting Clifford map does not become a projector simply because a local rank score is smaller. Either establish a valid adapter and charge it or keep that result in a separate primitive model. [S8, S21]

First discriminator: construct a complete small input/output/dirty word, including centers and boundary operations, then compute its total rank deficit and full characteristic. If every child strictly contracts and $H(0) \geq 1$, that paid profile cannot provide a positive saving. Preserve that obstruction and change the topology or representation; do not increase the same sweep merely to keep processors busy.

E5. Joint binary–complex progress toward 2×10^{-4}

The current public construction makes the earlier 10^{-4} target a useful comparison point rather than an unachieved community threshold. A meaningful next structural objective is an architecture that can support both a stronger binary primitive and a stronger complex primitive, or a rigorously different transfer that avoids the frozen coupling.

Run different hypotheses concurrently: a new odd-weight or center family; complete cancellation-enabled synthesis; a paid noncommuting or spectral interface; and an independent obstruction/formalization track. Give a normal computational job about four workers and permit oversubscription when memory allows. A track may read, derive or analyze while its processors are idle; distinct scientific questions matter more than CPU occupancy.

Do not reject a valuable bit component merely because the current complex supplier cannot exploit it yet. Conversely, do not report the component's saving as the final κ. Maintain a dependency ledger: which bit supplier, complex supplier, conversion, precision bound, routing primitive and recovery theorem would make the target possible? Necessary inequalities should expose impossible combinations before large searches.

General circuit results show that allowing cancellation can enlarge the available synthesis space, but they do not establish a gain for this particular framed map. Harvey–van der Hoeven's analytic framework and original circuit-complexity literature are appropriate primary-source inputs; the decisive deliverable remains a paid, compatible construction or a precisely scoped obstruction. [S25, S26]

7. E6: make the new interfaces more trustworthy

The highest scores inherit more than the original manuscript. The stopped-product family adds opposite-bank normalization, atom streaming, ordinary conversion and an exact odd-denominator grid. A complete finite replay can validate its finite input while leaving those all-size interfaces unproved. This distinction is explicit in #117's proof. [S3, S8–S10]

Highest-value formalization targets

Target	Required statement and boundary
Full residual factorization	Show the complete involution equals the proposed reversed child plus legal triangular adapters; include both diagonal blocks and complementary orientation.
Uniform stopped recurrence	Prove a bound uniform in atom count n and width w, including node adapters, peeled remainders and ordinary leaves. Track the strict moment gaps.
Outer wrapper placement	Prove exact ordinary interchange and restored state with a constant number of calls outside the recursion. Avoid an uncharged per-node conversion.
Odd-grid preservation	Bind the divisor-21 exponent to literal scalar operations and recursion depth; show completed returns preserve the incoming grid and final recovery encoding.
Current-word reflection	Prove or independently replay inverse signs, bank swaps, complemented frames, exact target readouts and arbitrary dirty restoration for the selected deferrals.

Our Lean package already formalizes the finite positive-matrix obstruction and its finite-level extension; it does not cover these new constructions. PR #64 offers much broader characteristic and machine-interface work, but for older pinned profiles and with remaining scheduler/routing obligations. Its 298 theorem audits do not automatically certify #120. Formalize the needed statement, not an impressive but unrelated theorem count. [S21, S22]

Verifier hardening is part of the research

PR #101 demonstrates why green checks require interpretation: earlier shared entry points could accept uncharged cancelling scatter gates, negative frame-index aliases and assertion-disabled Python. Its repair binds the exact paid incidence multiset and rejects those execution modes. This identifies classes of tests to include in independent current-word audits; it does not prove that every older submitted word was wrong. [S23]

Add deliberate mutations independently to each verifier: remove a cleanup, add an algebraically cancelling but unpaid operation, flip the signed correction, omit a frame complement, shorten a target chain, alter a mass-preserving histogram, and change the source pin. A validator that merely re-reads its own saved PASS flag cannot serve as independent reproduction.

8. Compatibility map and proposed allocation

The following matrix separates plausible integrations from combinations that should not be assumed valid. It is a research plan; no experiment in this table was launched on either server for this report.

Combination	Assessment	First decisive gate
#117 + latest #114 subspace selection	Already integrated by #120	Reproduce the baseline; do not claim this port as a new experiment.
#120 + joint or alternative-complement placement	Highest-priority next search	True full-profile improvement, not just more deferred dimensions.
#120 + #112 signed reclamation	Plausible, substantial integration risk	Signed common-frame synthesis and all dirty readout costs.
#120 + arbitrary #115 partial gauges	Not automatically additive	Check whether it duplicates existing deferral and whether all source/target chains remain legal.
One-child ranks + old coordinate-only search	Usually no direct gain	Demonstrate an actual changed rank list, schedule or legal selection; similarity alone preserves rank.
RaD Clifford / five-subset + #104	Structural research, not plug-in compatible	Prove the new residual interface and preserve the complete paid deficit.

Near-term GPU-host portfolio

Assign a baseline/reflection lane and independent E1 variants to the host’s CPU cores. Use the two GPUs for batched subspace/rank screening or clearly parallel candidate evaluation only after a measured implementation demonstrates useful throughput. Exact GF(2) work on 24-dimensional masks may be faster on the CPU than through a new GPU pipeline; accelerator utilization is not an acceptance metric.

Breakthrough CPU portfolio

Keep four or more distinct hypotheses where useful, normally about four CPU workers per computational job, without a three-job hard ceiling. Reserve memory for the OS and coordinator; limit aggregate memory and output growth rather than forcing a nominal utilization percentage. Preserve own task directories and one coordinator responsible for Git. These recommendations retain the user’s independence requirement: neither host needs direct messages or writable files from the other.

Suggested decision sequence

First: establish #117, #118 and #120 as immutable reference inputs with distinct evidence scope. Second: perform E1’s variants beyond #120’s integrated policy. Third: expand to E2/E3 only after exact scoring identifies a benefit. In parallel, leave the broad CPU research free to change geometry and interfaces rather than chase every update. These phases are evidence gates, not fixed completion promises.

9. What to record—and when to publish

One candidate identity, one complete evidence chain

Every experiment should bind the scalar DAG, selected frame/gauge policy, literal program, source hashes, complete child histogram, binary and complex suppliers, stopping parameters and assembly. Keep each changed candidate in a separate immutable run. Never mix one word's role count with another word's histogram or retain an old data-family count after changing dimensions.

Evidence stage	Minimum retained result
Discovery	Explicit changed hypothesis, configuration, seed and fast-screen score; no theorem promotion.
Exact finite result	Source/target identities, paid capacities and chronology, nondegeneracy as needed, literal dirty replay, all signed/reflected transitions and exact rank accounting.
Conditional exponent	Complete moments with rigorous enclosures, all relevant scalar/precision/routing/row/setup/recovery costs, and the stated transfer assumptions.
Reproduction	Fresh bounded regeneration from pinned inputs; separate evidence for full repository reruns and for code inherited without rerun.
Formal coverage	Named theorem statements and their actual source bindings, audited axioms and explicit exclusions.

For cheap comparative screening, record H(s) at a shared trial saving; for final acceptance, recompute the characteristic and all strict margins. Preserve rank mass, maximum child, number of deferred roles, readout expansion, wall time, peak memory and worker settings. A fixed number of extra gates can matter for precision and eventual thresholds even when it does not change the asymptotic saving.

Public communication should match the contribution

A structural improvement, independent repair or meaningful theorem can warrant publication even if another submission has a larger κ. A microscopic parameter adjustment to an unchanged network often belongs in a note on its parent PR. The numerical ranking is only one axis of scientific value. Explain separately whether the contribution changes the construction, improves its certificate, or strengthens its verification.

Once a complete accepted checkpoint exists, preserve its compact proof, source pins, exact certificate and reproduction recipe immediately. Do not delay this for large historical search archives. Before claiming a current record, refresh the leading PR heads and compare like-for-like scope. A source update can invalidate yesterday's comparison without invalidating the earlier certificate.

> No new κ has been discovered or certified for RaD in this review. The independent result here is the rational check of the supplied #118 and #120 complex moments. Their physical derivation and all-size assumptions were not validated by that calculation. New variants remain experiments until their complete evidence chain is reconstructed.

10. Avoid repeating exhausted or mismatched searches

The old ceiling was scoped, not universal

The earlier frozen complex root near 7.174462×10^{-5} and the old $\beta = 1/20$ assembly restriction remain valid only for their fixed inputs and assumptions. The new family changes both the complex profile and the compiler/transfer interfaces, and uses $\beta = 10^{-6}$ in the inspected certificate. Its larger scores therefore do not refute the old arithmetic obstruction; they leave its scope. [S8–S11, S21]

Do not carry an obsolete optimization objective forward

Under the old ordered-corner compiler, changing a coordinate flag could change how a matrix decomposed into contiguous recursive blocks. Under a proved one-child-per-projector rule, a rank-preserving relabeling of the same residual list leaves the child widths unchanged. A basis or ordering experiment is useful only if it changes legal physical choices, the residual ranks or multiplicities, admissible gauges, a different implementation cost, or the transfer itself. [S8, S19, S20]

Likewise, fewer roles are not automatically better. #114 improves its subspace-refinement profile without changing the graph, W or rank mass. #116 reports that a plausible combination of an apparently smaller producer and deferral lost. Evaluate the complete objective and retain counterexamples to naive proxies. [S4, S5]

Repeated types do not manufacture better primitives

The RaD positive-mixing argument rules out strict positive-potential contraction when every row of the unchanged complete moment matrix sums to at least one; the formal package also covers finite products of such levels. It does not rule out shared computation between child occurrences, new signed or geometric primitives, changed child sizes or a different paid recurrence. Repeating mixing-fraction sweeps inside the already excluded model is low-value. [S21]

Preserve the distinction between model changes and bugs

Changing the algorithm can justify a different ledger. Removing an inconvenient cost without a replacement implementation cannot. This is especially important for nonlinear CRT versus known-bit routing, exact versus rounded child returns, free reversal, zero-initialized versus arbitrary dirty scratch, and local Clifford gains versus complete-network chronology. The new CPU branch already records several of these distinctions; consume its summaries before opening another equivalent search. [S21, S22]

For this project, useful progress is a new compatible primitive, a demonstrable full-profile gain, a proof that closes an important interface, or an obstruction that meaningfully redirects research. A million variants in a frozen family may be less valuable than one small exact counterexample to a promising but unpaid transformation.

Appendix A. Pinned inputs and reproducible arithmetic

These pins identify the reviewed versions. A PR may subsequently change title, graph, κ or validation status. Never combine a current PR body with an older certificate without checking the source identities.

Input	Immutable revision
Reviewed main	56b66d58297deca1d7dd130247d720e960f77a37
#120 / eumemic	bfc5466b028923a1f8655602994ea56c70b5c329
#118 / rohanarun	ec862a5af51537495c745d9f2323e8a5736ee265
#117 / eumemic	cbb05ce504d571546d9b7794c186a613c659c3bf
#116 / rohanarun	97750ec74fc714638c0aa4f31aaa880e56cb8b4d
#114 / eumemic	5633a22443940f424dc2a09f44b2f447890e0074
#104 / icekylinx	948ce1510df750f4c18b96bdaef436a86f8bf834
#101 / Bortlesboat	76f55061bb50d8244c985807909a5f44254bd906
#64 / rfu08	2d970ab3ca0271cfb2b609dc9aae1b7073b7774d
RaD breakthrough reference	8135f5e4d3f2dbcd36bb2a52df341f4871b1e974

Companion arithmetic

The accompanying arithmetic archive contains check_profile.py, independent_arithmetic_final.json and explicitly labeled transcriptions of the #118 and #120 complex profiles. It requires only Python's standard library and performs no network access. Run it in its extracted directory:

```
python3 check_profile.py
```

It checks the rank identity and both acceptance/next-grid inequalities for each profile, then records the exact rational bounds and the derived headroom comparisons. It does not invoke a producer, the 47-row assembly checker, a Lean package or a remote service. No network-wide or all-size conclusion follows from this arithmetic alone.

Suggested source reconstruction—not executed here

#117 documents make deferred-product-verify. #120 documents make deferred-replayed-verify, which exports the word, replays the two stages with mutation controls, and checks the exact network certificate. Full make verify is a separate broader run. Acquire each pinned version with its dependencies. #114's complement method is already part of #120; no second certificate should be added. [S3, S13, S27]

Historical broad suites can be expensive and share regenerated files. Follow each package's sequential or isolated-checkout instructions; do not infer that make -jN is safe because the research host has N cores. A reported full test run must identify the exact source version, completed groups, source drift and any interrupted-and-resumed stages.

Appendix B. Sources and attribution

References identify original authors and exact sources where available. PR-page claims and statuses are a snapshot, not immutable evidence. Full revisions for the critical inputs are in Appendix A. Bibliographic links are clickable.

[S1] Reviewed main: selected result and review boundary. CrocSwap/integer-mult-bounds, README and main 56b66d58; submissions after #62 outside this review checkpoint. Open source

[S2] PR #118 — deferred readouts on the #117 producer. Rohan Arun; head ec862a5a. Numerical claim, 3,020 deferrals and pending full/reflected checks. Open source

[S3] PR #117 — 28,705-role complex producer. eumemic; head cbb05ce5. Non-draft submission and author-reported full Linux reproduction. Open source

[S4] PR #116 — cube recurrence with balanced deferred frames. Rohan Arun; head 97750ec7. Compare its older #114 reference against the updated #114 explicitly. Open source

[S5] PR #114 — updated core-aware cyclic deferred network. eumemic; head 5633a224. Radical-complement refinement, signed reflection audit and scalar charge. Open source

[S6] PR #104 — stopped product-ring interchange. icekylinx; head 948ce151. New compilation, atom, rational-center and semantic interfaces. Open source

[S7] PR #97 — deferred endpoints and signed phase correction. Zhihao Chen / jacklightChen, using Swapnil Jain's frozen round-seven inputs. Physical and signed endpoint integration. Open source

[S8] #104: complete opposite-bank factorization. Pinned mathematical source: common basis, full involution and the one-reversed-child construction. Open source

[S9] #104: paid atom recurrence and ordinary wrapper. Pinned mathematical source: complete tolls, stopping conversion and uniformity claims. Open source

[S10] #117: proof and explicit dependency scope. Nested bit residual inventory, new complex producer, reflected checks and retained all-size assumptions. Open source

[S11] #118: supplied complex profile and exact certificate driver. Source of the independent arithmetic input and constants; companion certificate.py in the same pinned directory. Open source

[S12] #118: actual candidate-selection implementation. Pinned complex_deferred.py, especially the deferral-frame admission and nesting loop. Open source

[S13] #114: nondegenerate complement and placement code. Pinned complex_deferred.py: sat_nonsingular_part and saturated_placement. Open source

[S14] PR #115 — partial source gauges and cube producers. icekylinx. Focused finite validation, backward frame enlargement and source-gauge chronology. Open source

[S15] PRs #110, #111 and #113 — deferral and cyclic-strip lineage. Avi Eisenberg (#110) and Rohan Arun (#111/#113); read each at its own proof scope. #113 includes the expanded scalar-charge correction. Open source

Sources and attribution (continued)

Additional source context and research boundaries. Primary source descriptions were used for the comparisons; this report does not claim to have rerun their full suites.

[S16] PR #112 — signed reclaimed complex supplier. Finite signed dependence reclamation; final κ limited by its retained binary supplier. Open source

[S17] PR #109 — weighted dual-suffix carriers. Rohan Arun. Same role count with different paid widths; author-reported full verification. Open source

[S18] PR #74 — a mutable proposal with a newer research family. Thomas DiFiore; inspected body identifies current head b79fbc4b and stopped pair-star trees, not its early aligned-frame score. Open source

[S19] PR #91 — aligned-frame composition and scoped Lean arithmetic. Formal arithmetic does not formalize physical input derivation or the full multiplication theorem. Open source

[S20] PR #106 — contiguous-pair pricing and circuit exchange. An incremental old-track result, not the global numerical leader. Open source

[S21] RaD breakthrough research: current components and obstructions. Reference head 8135f5e4; read the linked individual reviews before treating a component as an accepted interface. Open source

[S22] PR #64 — finite profiles and machine-transfer formalization. rfu08; head 2d970ab3. Stated formal coverage and explicit routing/scheduler/analytic limitations. Open source

[S23] PR #101 — reject unsound binary-frame verifier inputs. Bortlesboat; head 76f55061. Exact paid scatter binding, negative-index rejection and optimized-Python controls. Open source

[S24] PR #96 — merged auxiliary exteriors. eumemic. Historical example of changing full residual profiles and maximum-child/row-stock requirements without changing the word's roles. Open source

[S25] Harvey and van der Hoeven, Integer multiplication in time O(n log n). Annals of Mathematics 193 (2021), 563–617. Primary analytic background, not validation of the new community interfaces. Open source

[S26] Boyar and Find, Cancellation-Free Circuits in Unbounded and Bounded Depth. arXiv:1305.3041. General circuit-complexity context; its separations do not establish a gain for the present framed map. Open source

[S27] PR #120 — final-refresh leader with saturated placement. eumemic; head bfc5466b. Already combines #117 and #114; finite reflected-word audit reported, full suite pending. Open source

[S28] PR #119 — closed, unmerged partial-complex study. seanabreau; head cf46d090. Signed adjoint/access-order accounting and scoped finite Lean arithmetic; not an open leader. Open source

[S29] #120: complete supplied complex histogram. Source of the second independent rational check; W, rank and maximum child are unchanged from #118 while the histogram changes. Open source

For [S15], the companion PR pages are #111 (cyclic strips on both axes) and #113 (deferred cyclic network and scalar-accounting correction).

PR #111 • PR #113

The main implementation lineage retains attribution to Swapnil Jain, Zhihao Chen, icekylinx, Avi Eisenberg, eumemic, Rohan Arun, Rohan Gupta, Chafik Boukhalfa, Aurel Prosz, James Chang, RaD/hipotures and the contributors named by each imported source. Douglas Colkitt maintains the community repository; OpenAI and Harvey–van der Hoeven retain their respective manuscript and analytic contributions. This review makes no novelty claim for imported methods.

Prepared with AI assistance. Source inspection and the independent arithmetic check are described explicitly; proposed experiments are not completed research results.